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Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Integrated Systems Group		
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Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
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Zip Code:	47907-2010	**Congressional District:**	4
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	The principal investigators, postdoctoral staff, graduate students, and technical staff have continued to engage in several projects and events. A dynamic human model and a crew cabin atmosphere model have been built. A prototype SIMulation based OPTimization (SIMOPT) framework is being used to enable identification of the optimal strategies to control the overall ALSS behavior. A Plant-Scheduling model has been developed to determine the best planting schedules to minimize ESM. ESM analysis is being conducted on all subsystems, and has initial computations have been completed on several subsystems – UV water disinfection, BREATHe II, STAR, LiFT, Zeolite, biomass production chambers and food production subsystems. ESM sensitivity analysis is also being conducted. A process model of all mass flows in a life support system relying uniquely on proposed NSCORT technologies has also been developed. A system dynamics model using Stella software was developed for the Membrane System for ALS Wastewater Recycle. Another prototype high-level 'qualitative' system dynamics model of a complete bioregenerative life-support system was also built using Stella. These models are also being incorporated into the senior level Systems Management Course at Howard University. Near-term future work will involve continued interaction with the members of all the NSCORT systems to refine and extend the models being developed and the ESM calculations.
Task Description:	Guy Gardner Integrated Systems Group – Focus Area Lead For Systems Modeling of ALS: • Develop a SIMulation based OPTimization (SIMOPT) framework to study the dynamics of advanced life support system. Evaluate ESM of technologies developed in ALS NSCORT. Determine the technology performance by comparing with the ISS assembly complete technologies ESM. • Develop a dynamic model to investigate the overall system performance. For a System dynamics approach to modeling the ALS system: • One objective of this project is to identify and apply critical developments in knowledge management to the knowledge engineering tasks of the overall NSCORT effort. This emphasis of knowledge management is designed to facilitate ESM identification and reduction. • The second broad objective focusing on the application of the system dynamics methodology to the study and analysis of the Advanced Life Support (ALS) system as a whole as well as its subsystems. This approach is designed to provide both conceptual models and quantitative computer simulation models that will facilitate further understanding of the dynamics involved in the ALS system
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of computer models for control of complex processes
	ALS/NSCORT systems group research activities in its second year are aligned with the November 2003 External Advisory Committee meeting recommendations and can be grouped in three major categories: Model Development Activities, ALS/NSCORT Process Map Development and ESM estimations of ALS/NSCORT technologies. Model Development Activities A dynamic human model is built to evaluate the crew members' inputs and outputs by hour. In this model, users can freely assign crewmembers' gender, age, weight, and activity schedules to find out the hourly energy expenditure, heat production, oxygen requirement, carbon dioxide and respiration water productions, as well as the diet composition. Waste generation is computed in proportion of the total oxygen, food and potable water consumed. Meanwhile, hygiene water inputs and outputs are extracted directly from Baseline Values and Assumptions Documents, 2002. Some of the data and formula applied are not as sensitive as others with respect to crewmember's gender, age, weight and activity schedules (e.g. hygiene water consumption). This is mainly due to the lack of existing formula and data available. However, this shortcoming is not a hurdle in developing the dynamic model, as the use of Excel spreadsheet in computing these attributes allows users to easily update formula and data later. Apart from the human model, crew cabin atmosphere is modeled as well. In the crew cabin atmosphere, the habitat volume, total cabin pressure, air temperature and mass of oxygen, carbon dioxide, water vapor, and nitrogen by difference are considered. Using Matlab, these two subsystems are combined together to evaluate the effects of crewmembers' activities toward the cabin atmosphere. To this date, the oxygen depletion and increase of carbon dioxide, water vapor and temperature in the cabin are successfully modeled. A SIMulation based OPTimization (SIMOPT) approach is used to study the dynamics of Advanced Life Support System (ALSS). The SIMOPT architecture uses a Deterministic Optimization (DO) algorithm to optimize the overall ALS behavior by minimizing the re-supplies which are difficult to procure or transport, in conjunction with a simulation model which introduces uncertainty, i.e. randomness, to the system. A prototype SIMOPT framework is studied. For this purpose, a simple dynamic simulation of the ALSS using an aggregate mass balance model is developed to represent reality with uncertainty. The ranges of acceptable values of strategic decisions, e.g. safety buffer for oxygen, in a given ALSS scenario are determined using time series data mining methods performed on the information gathered within the timelines, which are generated by the ALSS simulation. These buffer values impose constraints to the optimization model (DO algorithm) which is an aggregate steady state mass balance model. This model determines the optimal values for the degrees of freedom of the ALSS used in the simulation. It should be noted that the random events introduced through the simulation may result in a difference between the predicted values of the variables by optimization model and values generated by the simulation (a trigger event). As the number of trigger events increases the data mining algorithm examines the timelines to infer the system behavior and updates the desired safety buffer values. These new values are used in the optimization model within the proposed SIMOPT architecture. Therefore, SIMOPT framework enables us to identify the optimal strategies to control the overall ALSS behavior. This work has been presented in International Conference on Environmental Systems (ICES) 2004. Additionally, a Plant-Scheduling module is developed. Given the demands of the ALSS crops over a period of time (which can be specified by the user of the module), it determines the best planting schedule that will optimize the system behavior, i.e. the one that would minimize ESM. The formulation is an MILP and coded in Java, the equations are solved using CPLEX, which is a widely used commercial optimization package. ALS/NSCORT Process Map Development A proccess model of all mass flows in a life support system relying uniquely on proposed NSCORT technologies is developed. The purpose of this activity was to answer the question whether the ALS/NSCORT technologies can act as a system or not (see appendix for the current ALS/NSCORT process map). The model has an Excel interface to unify all specifications of process connectivity, stream compositions, mass and concentration values, reaction stoichiometry, phase equilibria, and other relations. The underlying code solves a linear program currently having 947 variables and 711 equations. The model finds feasible solutions including operation of all air, water, and waste recovery processes, growth of 15 crops, edible fungi, tilapia, and six biosolids dewatering plants. The air and

water loops are evidently closed, and food is mainly supplied based on the current choice of costs for supply vs. growth. Biomass production is minimized in general. ESM estimations of ALS/NSCORT technologies ESM analysis is conducted to evaluate the performance of ALS/NSCORT technologies. As most of the technologies are still under development, appropriate assumptions and educated guesses from subsystem experts are considered into the analysis. This ESM analysis focuses on a hypothetical device, instead of the anticipated technology that is system flight proven in mission operations, because many ALS/NSCORT technologies are at low TRL levels. To add more flexibility into the ESM calculation, sensitivity analysis is introduced to include all the ESM possible values. In addition, this method allows the identification of the most critical factor that can reduce the ESM significantly. So far, the ESM evaluation on UV water disinfection, BREATHe II, STAR, LiFT, Zeolite, biomass production chambers and food production subsystems are completed. Initial ESM estimation and sensitivity analysis for the NSCORT ALS subsystems are performed for Mars surface mission scenario. The estimations based on BVAD requirement and consider the physical/chemical constraints of each subsystem as well as preliminary experimental data provided by the subsystem development team. Sensitivity analysis of the estimated ESM provided information for the subsystem development team to optimize system design and reduce ESM for the entire advance life support system. Moreover, the established mission scenarios documented in the RMD are used to analyze the impact of mission scenario on subsystem design for certain ALS/NSCORT technologies. The purpose of this analysis is to explore the impact of the various ESM equivalencies of different missions. Future Research Directions The ALS/NSCORT systems group envisions continuing to work effectively in these three major activity threads in the upcoming year. Model Development Activities After building the human and cabin atmosphere model, crop model will be constructed and then incorporated with the two models mentioned. At this point of time, the modified energy cascade available in BVAD 2004 is selected as a “candidate” to model the crop. It is anticipated that the crop model will be able to capture the daily oxygen production, daily carbon gain, daily canopy transpiration rate, total biomass and edible biomass, among others. For simplicity purpose, a hypothetical crop is considered at this time. The hypothetical crop will possess the average properties of dry bean, lettuce, peanut, rice, soybean, sweet potato, tomato, wheat and white potato, which are the crops that have the most complete data for use in the modified energy cascade. Currently a diet optimization module which will determine the best diet cycle that minimizes the ESM of the overall ALSS is being developed. Given the activity schedule of the crew members, the model will be able to calculate the necessary nutritional requirements of the crew members using a dynamic human model. A diet cycle (20-30 days cycle) will be constructed that meets these requirements while minimizing the ESM. This model will also be able to give the necessary biomass amount required to construct this diet. This study will be completed by the end of 2004.

Task Progress:

Necessary biomass amounts calculated by this module will be fed into the plant-scheduling module as the demands. The diet optimization and plant-scheduling modules will be integrated into the SIMOPT framework to study their effects on the overall system behavior and to represent the ALSS more realistically. We will analyze the supply requirements and waste generations of three different scenarios, i.e. camping trip approach in which everything that is need for life support is shipped to Mars, a more evolved base including pysico-chemical systems for resource recovery and finally a complete base with biomass production via the plant growth chambers, for Mars surface missions. The system trade of studies will be done on the architecture level using ESM values as well as in terms of waste generation by the system. This work will be submitted as a manuscript to a special issue of International Journal of Environment and Pollution (IJEP) in January 2005. ALS/NSCORT Process Map Development The process map developed will be a living effort which will be updated and modified as more detailed data and information is evolved from ALS/NSCORT projects to investigate integration issues such as, but limited to, load compatibility, transient behavior, and fault/error handling. The result of the investigation will provide sub-system developer more reliable information on interconnect behavior and fault/exception handling requirements. This analysis will be able to provide feedback to the sub-system researchers regarding system integration and cost trade-offs and ensures that the sub-system will be able to function in an integrated life support system. Sensitivity analysis is can be applied to the ESM result to identify aspect of the system that will provide most significant improvement. ESM estimations of ALS/NSCORT technologies ESM analysis will also be conducted on BREATHe I, iodine water disinfection, reverse osmosis/ nano filtration, biosolids dewatering plants, fish/tilapia and edible fungi growth subsystems. After that, the performances of each subsystem will be evaluated through comparison with the ISS assembly complete technologies. As ISS assembly complete technologies are expected to have lower closure or higher re-supply than ALS technologies, not all ALS NSCORT technologies will be able to find a compatible ISS subsystem to compare its performance with. This challenge remains to be addressed. The ESM of a complete ALS system will be estimated after a mission scenario specific NSCORT ALS design has been finalized. The total system level ESM estimation, in addition to subsystem ESM, needs to include system integration, interface and control functionalities. The result will be compared to the ISS or other similar ALS systems.

• System integration: We will begin to integrate individual NSCORT ALS technologies and investigate integration issues such as, but limited to, load compatibility, transient behavior, and fault/error handling. The result of the investigation will provide sub-system developer more reliable information on interconnect behavior and fault/exception handling requirements. This analysis will be able to provide feedback to the sub-system researchers regarding system integration and cost trade-offs and ensures that the sub-system will be able to function in an integrated life support system. Sensitivity analysis is can be applied to the ESM result to identify aspect of the system that will provide most significant improvement. • ESM estimation for NSCORT ALS: The ESM of a complete ALS system will be estimated after a mission scenario specific NSCORT ALS design has been finalized. The total system level ESM estimation, in addition to subsystem ESM, needs to include system integration, interface and control functionalities. The result will be compared to the ISS or other similar ALS systems. Lunar space missions have been proposed to be the first to investigate.

A dynamic human model is built to evaluate the crew members’ inputs and outputs by hour. In this model, users can freely assign crewmembers’ gender, age, weight, and activity schedules to find out the hourly energy expenditure, heat production, oxygen requirement, carbon dioxide and respiration water productions, as well as the diet composition. Waste generation is computed in proportion of the total oxygen, food and potable water consumed. Meanwhile, hygiene water inputs and outputs are extracted directly from Baseline Values and Assumptions Documents, 2002. Some of the data and formula applied are not as sensitive as others with respect to crewmember’s gender, age, weight and activity schedules (e.g. hygiene water consumption). This is mainly due to the lack of existing formula and data available. However, this shortcoming is not a hurdle in developing the dynamic model, as the use of Excel spreadsheet in computing these attributes allows users to easily update formula and data later. Apart from the human model, crew cabin atmosphere is modeled as well. In the crew cabin atmosphere, the habitat volume, total cabin pressure, air temperature and mass of oxygen, carbon dioxide, water vapor, and nitrogen by difference are considered. Using Matlab, these two subsystems are combined together to evaluate the effects of crewmembers’ activities toward the cabin atmosphere. To this date, the oxygen depletion and increase of carbon dioxide, water vapor and temperature in the cabin are

	successfully modeled. ESM analysis is conducted to evaluate the performance of NSCORT technologies. As most of the technologies are still under development, appropriate assumptions and educated guesses from subsystem experts are considered into the analysis. This ESM analysis focuses on a hypothetical device, instead of the anticipated technology that is system flight proven in mission operations. To add more flexibility into the ESM calculation, sensitivity analysis is introduced to include all the ESM possible values. In addition, this method allows the identification of the most critical factor that can reduce the ESM significantly. So far, the ESM evaluation on UV water disinfection and BREATHe II subsystems are completed. Charita Brent developed a high level membrane system incorporating all membrane subsystem 0th order parameters. This was done using a system dynamics approach and Stella software She set up scenarios for each RO/NF membrane (6) selected by the Membrane Group using projected rejection rates at maximum pressure levels. She ran simulations using different backwash intervals and experimented with various recursive algorithms to match subsystem water recovery activity. Different knowledge elicitation processing exercises were developed to archive project knowledge. A combination of techniques including interviews, task analysis and process tracing were used. John Trimble developed a prototype high-level 'qualitative' system dynamics model based on the "bioregenerative life-support system (modified from Hoff et al. 1982)" that is presented in the Introduction of our research plan description. This model is built using Stella. It will be used to illustrate the general causal relationships between components of the ALS system and develop a better understanding of the precise quantitative relationships in play as our complex system evolves. John Trimble has modified the Systems and Computer Science senior level Systems Management Course to include components that utilize the system dynamics models build for the NSCORT project. The course will use both the 'Water subsystem' models developed by Dr. Trimble and extended by Charita Brent over the past year and the recently developed 'prototype high-level' model. This course is a required course of all systems and computer science majors at Howard University and presents an opportunity to involve a wider range of students in our project and also identify potential graduate students with an interest in the systems component of ALS. Interviews of the Howard University investigators were conducted to access the utilization and usefulness of the ALS-NSCORT web portal. These interviews and discussion with other team members is the basis of the development of a survey questionnaire to be completed by all faculty and student investigators to further access the web portal usefulness and utilization. The intent is to determine ways to make the portal and the associated project communication processes more effective. Work over the summer was conducted by students in conjunction with Dr. Keeling to investigate semantic web developments and related tools. This effort is designed to provide insights into how to incorporate the semantic web capabilities into a future web-based knowledge repository for the project.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Papers from Meeting Proceedings	Aydogan S, Orcun S, Blau G, Pekny JF, Reklaitis G. "A Prototype Simulation Based Optimization Approach to Model and Design an Advanced Life Support System." International Conference on Environmental Systems, Colorado Springs, CO, July 2004. International Conference on Environmental Systems, July 2004. Paper 2004-01-2576. , Jul-2004
Presentation	Selen Aydogan, Seza Orcun, Gary Blau, Joseph F. Pekny and Gintaras Reklaitis "A Prototype Simulation Based Optimization Approach to Model and Design an Advanced Life Support System" Poster Presentation, Habitation 2004 Jan-2004
Presentation	Brent, Charita, Trimble, J, "Applying Knowledge elicitation Techniques to Construct Membrane Flight " National Technical Association Annual Conference Sep-2004
Presentation	Brent, Charita "Applying Knowledge Elicitation Techniques to Construct Membrane Flight Simulator" poster at the International Multi-conference in Computer Science and Computer Engineering Las Vegas, Nevada . June 2004 Jun-2004
Presentation	George Applequist, Joseph F. Pekny, Seza Orcun, Selen Aydogan, Gary Blau, George Chiu and Bin Yao "An Aggregate System Level Mass and Energy Optimization Model of an Advanced Life Support System" Poster Presentation, Habitation 2004, (2004). Jan-2004